

Statistics

2025 Lectures

Part 5 - Conditional Distributions

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Recommended reading

For this lecture (Part 5), please, read

- sections 6.3, 7.7 from Bartoszynski and Niewiadomska-Bugaj (11 pages)
- sections 2.6, 3.5 from Mittelhammer (12 pages)

Conditional distributions

- so far, no additional information has been available relating to the occurrence of an event for any subset of the random variables $(X_1, \dots, X_n)$

Example 56: The expected number of housing starts (i.e. the number of new residential construction projects initiated within a given month), calculated for planning purposes by building supply manufacturer, may depend on an assumed level of mortgage interest rates.

- conditional distributions and conditional expectation are central to regression analysis
- relates to an experiment where n different characteristics are recorded for each outcome. We are specifically interested in analyzing a subset of these characteristics, given that certain values for the remaining characteristics are fixed with certainty.

Discrete conditional distributions

- In the bivariate case, the question is to determine

$P(X \in Q_1|Y = y)$ or $P(Y \in Q_2|X = x)$

a) discrete case

- if $P(Y = y_j) = 0$ then we leave the conditional probability undefined
- if $P(Y = y_j) > 0$ then

$$P(X \in Q_1|Y = y_j) = \frac{P(X \in Q_1, Y = y_j)}{P(Y = y_j)} = \frac{\sum_{i:x_i \in Q_1} p_{ij}}{\sum_i p_{ij}} = \frac{\sum_{i:x_i \in Q_1} p_{ij}}{p_{\cdot j}}$$

- analogously,

$$P(Y \in Q_2|X = x_i) = \frac{\sum_{j:y_j \in Q_2} p_{ij}}{\sum_j p_{ij}} = \frac{\sum_{j:y_j \in Q_2} p_{ij}}{p_{i\cdot}}$$

Example 57: Let X_1 and X_2 denote the result of the first and second toss of a die and put $Z = |X_1 - X_2|$. Determine the distribution of $Z|X_1 = 5$.

Continuous conditional distributions

Definition 25: (conditional density)

The **conditional densities** g_{12} and g_{21} are defined by

$$g_{12}(x|y) = \frac{f(x, y)}{f_Y(y)}, \text{ provided } f_Y(y) > 0,$$

$$g_{21}(y|x) = \frac{f(x, y)}{f_X(x)}, \text{ provided } f_X(x) > 0,$$

where f is the joint density of (X, Y) and f_X, f_Y are the marginal densities.

Note that conditional densities are indeed valid density functions:

- nonnegative in the first variable

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$$\int_{-\infty}^{\infty} g_{12}(x|y) \mathrm{d}x = \frac{1}{f_Y(y)} \int_{-\infty}^{\infty} f(x, y) \mathrm{d}x = \frac{f_Y(y)}{f_Y(y)} = 1.$$

Conditional distributions and thinning process

Example 58: Let the joint density of (X, Y) be given by

$$f(x, y) = \begin{cases} cxy^2, & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \geq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Determine the conditional distribution of X given $Y = 0.5$.

Example 59: An animal lays a random number Y of eggs, where Y follows a Poisson distribution with parameter $\lambda > 0$. Each egg hatches independently with probability $0 < p < 1$. Determine the distribution of the number of eggs X that hatch.

- This scenario can be understood as a **thinning process**, where some of the Y objects disappear (are not counted) due to a random selection process. In this case, the thinning process is called **binomial thinning**.

Mixture of distributions

Example 60: Assume Y takes $n \geq 2$ values with $P(Y = y_i) = p_i$, with $\sum_i p_i = 1$. Next, assume that for a given $Y = y_i$, the random variable X has a continuous distribution with density φ_i . Thus,

$$P(a \leq X \leq b | Y = y_i) = \int_a^b \varphi_i(x) dx.$$

Apply Total probability formula to find the marginal (unconditional) distribution of X :

$$P(a \leq X \leq b) = \sum_{i=1}^n P(a \leq X \leq b | Y = y_i) P(Y = y_i) = \int_a^b \sum_{i=1}^n p_i \varphi_i(x) dx$$

where

$$f_X(x) = \sum_{i=1}^n p_i \varphi_i(x)$$

is called the **mixture of densities** φ_i with mixing coefficients p_i .

Conditional cdf and independence

- one can define analogously to the unconditional case the **conditional cumulative distribution function**. E.g. in the bivariate case:

$$F_X(t|Y \in Q_2) = P(X \leq t|Y \in Q_2).$$

- naturally, in the case of X and Y independent
 - discrete case:

$$P(X = x_i|Y = y_j) = P(X = x_i)$$

- continuous case

$$g_{12}(x|y) = f_X(x)$$

$$g_{21}(y|x) = f_Y(y)$$

Conditional expectation

Definition 26: (conditional expectation)

For random variables X and Y , the **conditional expectation** $E(X|Y)$ is defined as the random variable with values $E(X|Y = y)$ when $Y = y$.

More generally, for a real-valued function g ,

- in the discrete case, the values

$$E(g(X)|Y = y_j) = \sum_i g(x_i)P(X = x_i|Y = y_j)$$

occurs with probability $P(Y = y_j)$.

- in the continuous case, the values

$$E(g(X)|Y = y) = \int_{-\infty}^{\infty} g(x)g_{12}(x|y)\mathrm{d}x, y \in \mathbb{R},$$

are distributed with respect to $f_Y(y), y \in \mathbb{R}$.

Iterated expectation

Example 59 (cont.): Recall, $Y \sim POI(\lambda)$ and $X|Y = n \sim BIN(n, p)$. This means that the expected number of hatched eggs, provided $Y = n$, is $E(X|Y = n) = np$. So we have $E(X|Y) = Yp$.

Theorem 30: (iterated expectation theorem)

For any random variables X and Y and real-valued function g , we have

$$E(E(g(X)|Y)) = E(g(X)),$$

provided $E(g(X))$ exists.

Example 59 (cont.): Since Y has Poisson distribution with mean λ ,

$$E[E(X|Y)] = E(Yp) = E(Y)p = \lambda p.$$

This also follows from the fact that $X \sim POI(\lambda p)$.

Iterated variance

Theorem 31: (without proof) For any random variables X and Y , if $E(X^2)$ exists, then

$$\text{Var}X = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)].$$

Example 59 (cont.): As $X \sim \text{BIN}(Y, p)$ and $E(X|Y) = Yp$, $\text{Var}(X|Y) = Yp(1 - p)$; and $Y \sim \text{POI}(\lambda)$ and $E(Y) = \text{Var}Y = \lambda$. Thus

$$\text{Var}X = E[Yp(1 - p)] + \text{Var}(Yp) = \lambda p(1 - p) + \lambda p^2 = \lambda p.$$

This is in accordance with the fact that $X \sim \text{POI}(\lambda p)$.

Substitution theorem

In cases where one is conditioning on an elementary event $Y = y$, there are useful generalizations of the conditional formulas.

Theorem 32: (without proof)(substitution theorem)

For any random variables X and Y and a real-valued mapping g , we have

$$E[g(X, Y)|Y = y] = E[g(X, y)|Y = y].$$

Example 59 (cont.): Consider $Z = Y - X$ which denotes the number of unhatched eggs.

$$\begin{aligned} E(Z|Y = n) &= E(Y - X|Y = n) = E(n - X|Y = n) = \\ &= n - E(X|Y = n) = n(1 - p). \end{aligned}$$

In fact $Z|Y = n \sim \text{BIN}(n, 1 - p)$ and $Z \sim \text{POI}(\lambda(1 - p))$.

Recommended reading (week 7)

For the next lecture Part 6, please, read

- sections 8.1, 8.2, 8.3, 8.4, 8.5 from Bartoszynski and Niewiadomska-Bugaj (42 pages)
- sections 4.1, 4.2, 4.3, 4.4 from Mittelhammer